

# A LATTICE ENDPOINT SUBCASE FOR A QUESTION OF HYTONEN–VERAAR

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## STATUS

This packet gives a partial positive answer to the endpoint question after Lemma 3.1 in Hytonen–Veraar [1]. The source proves the estimate with  $\tilde{q} > q$  for Banach spaces of cotype  $q$ , and asks whether one can take  $\tilde{q} = q$  when  $q \neq 2$ . The result below proves the endpoint for a natural lattice class: Banach lattices with the Rademacher square-function equivalence and  $q$ -concavity.

## SOURCE PASSAGE

The source passage, on PDF page 7, reads that the authors do not know whether one can take  $\tilde{q} = q$  in Lemma 3.1(2) if  $q \neq 2$ , while noting the positive  $L^q$  case.

(3): Let  $x_1, \dots, x_N \in X$ . Let  $(S_n)_{n=1}^N$  be disjoint sets in  $\Sigma$  with  $\mu(S_n) = \mu(S_1) \in (0, \infty)$  for all  $n$ . Letting  $f_n = \mu(S_1)^{-1/q} \mathbf{1}_{S_n}$  for  $n = 1, 2, \dots, N$ , we obtain that

$$\left\| \sum_{n=1}^N r_n f_n x_n \right\|_{L^q(S; L^2(\Omega; X))}^q = \sum_{n=1}^N \|x_n\|^q.$$

On the other hand,

$$\mu(|f_n| > t) = \mu(S_1) \cdot \mathbf{1}_{[0, \mu(S_1)^{-1/q}]}(t)$$

for all  $n = 1, \dots, N$ . Therefore, (3.1) implies that

$$\sum_{n=1}^N \|x_n\|^q \leq C^q \left\| \sum_{n=1}^N r_n x_n \right\|_{L^2(\Omega; X)}^q,$$

which shows that  $X$  has cotype  $q$ . □

We do not know, whether we can take  $\tilde{q} = q$  in Lemma 3.1(2) if  $q \neq 2$ . However, if  $X$  is an  $L^q$ -space with  $q \geq 2$ , then one may take  $\tilde{q} = q$ . This follows from the next remark in the case that  $\tilde{X} = \mathbb{R}$ .

## DEFINITIONS

For a finite sequence  $(x_n)$  in a Banach space  $X$ , write

$$\|(x_n)\|_{\text{Rad}(X)} := \left( \mathbb{E} \left\| \sum_n r_n x_n \right\|_X^2 \right)^{1/2},$$

where  $(r_n)$  are independent Rademacher signs.

Let  $X$  be a Banach lattice. Say that  $X$  has square-function constant  $K$  if, for every finite sequence  $(x_n)$  in  $X$ ,

$$K^{-1} \left\| \left( \sum_n |x_n|^2 \right)^{1/2} \right\|_X \leq \|(x_n)\|_{\text{Rad}(X)} \leq K \left\| \left( \sum_n |x_n|^2 \right)^{1/2} \right\|_X.$$

We say that  $X$  is  $q$ -concave with constant  $M$  if, for all finite positive families  $(u_j)$  in  $X$ ,

$$\left( \sum_j \|u_j\|_X^q \right)^{1/q} \leq M \left\| \left( \sum_j u_j^q \right)^{1/q} \right\|_X.$$

The same inequality for simple positive  $X$ -valued functions follows by approximation.

## IDEA OF THE PROOF

The source proof loses the endpoint because the general cotype factorization lands in a Lorentz space. In a lattice with a square-function model for Rademacher sums, the endpoint estimate can be checked pointwise in the lattice variable. At each point it becomes the elementary scalar inequality

$$\left\| \left( \sum_n |a_n f_n|^2 \right)^{1/2} \right\|_{L^q(S)} \leq \left( \sum_n |a_n|^2 \|f_n\|_{L^q(S)}^2 \right)^{1/2},$$

valid because  $q \geq 2$ . The  $q$ -concavity of the lattice is precisely what allows the  $L^q(S)$  norm of the lattice norms to pass inside the lattice.

## PARTIAL THEOREM

**Theorem 1.** *Let  $2 \leq q < \infty$ , let  $(S, \Sigma, \mu)$  be a sigma-finite measure space, and let  $X$  be a Banach lattice with square-function Rademacher equivalence constant  $K$  and  $q$ -concavity constant  $M$ . Then there is a constant  $C \leq K^2 M$  such that for every finite family  $(f_n)$  in  $L^q(S)$  and every finite family  $(x_n)$  in  $X$ ,*

$$\left\| \sum_n r_n f_n x_n \right\|_{L^q(S; L^2(\Omega; X))} \leq C \sup_n \|f_n\|_{L^q(S)} \left\| \sum_n r_n x_n \right\|_{L^2(\Omega; X)}.$$

The lattice hypotheses imply that  $X$  has cotype  $q$ , so this is a genuine subcase of the source question. Thus the endpoint  $\tilde{q} = q$  in Lemma 3.1(2) of [1] holds for this lattice class.

*Proof.* By homogeneity, assume  $\sup_n \|f_n\|_{L^q(S)} \leq 1$ . For  $s \in S$ , the square-function equivalence gives

$$\left\| \sum_n r_n f_n(s) x_n \right\|_{L^2(\Omega; X)} \leq K \left\| \left( \sum_n |f_n(s)|^2 |x_n|^2 \right)^{1/2} \right\|_X.$$

Set

$$h_s = \left( \sum_n |f_n(s)|^2 |x_n|^2 \right)^{1/2} \in X_+.$$

Using  $q$ -concavity, first for simple functions and then by monotone/simple approximation, we obtain

$$\left( \int_S \|h_s\|_X^q d\mu(s) \right)^{1/q} \leq M \left\| \left( \int_S h_s^q d\mu(s) \right)^{1/q} \right\|_X.$$

Pointwise in the lattice variable, the scalar inequality  $L^q(S; \ell_2) \hookrightarrow \ell_2(L^q(S))$  for  $q \geq 2$  gives

$$\begin{aligned} \left( \int_S \left( \sum_n |f_n(s)|^2 |a_n|^2 \right)^{q/2} d\mu(s) \right)^{1/q} &\leq \left( \sum_n |a_n|^2 \|f_n\|_{L^q(S)}^2 \right)^{1/2} \\ &\leq \left( \sum_n |a_n|^2 \right)^{1/2}. \end{aligned}$$

Applying this with  $a_n = |x_n|$  in the lattice functional calculus yields

$$\left( \int_S h_s^q d\mu(s) \right)^{1/q} \leq \left( \sum_n |x_n|^2 \right)^{1/2}.$$

Therefore

$$\begin{aligned} \left\| \sum_n r_n f_n x_n \right\|_{L^q(S; L^2(\Omega; X))} &\leq KM \left\| \left( \sum_n |x_n|^2 \right)^{1/2} \right\|_X \\ &\leq K^2 M \left\| \sum_n r_n x_n \right\|_{L^2(\Omega; X)}. \end{aligned}$$

This is the desired endpoint estimate.  $\square$

## SCOPE AND LIMITATIONS

The theorem does not settle the full question for arbitrary Banach spaces of cotype  $q$ . Its hypotheses are lattice hypotheses, not consequences of cotype alone in the formulation used here. It does, however, push the endpoint beyond the explicitly mentioned  $L^q$  example in the source: the proof works for any lattice class where the square-function Rademacher equivalence and  $q$ -concavity constants are available.

## VERIFICATION NOTES

The proof is analytic. The script `code/make_open_problem_crop.py` was used only to generate the source-evidence crop. Cheap local indexes were searched for the arXiv id and core terms before promotion. A bounded web search on 2026-06-20 for the exact endpoint phrase,  $q$ -concave Banach lattices, and  $R$ -boundedness did not surface a direct duplicate.

## REFERENCES

- [1] T. Hytonen and M. Veraar, *R-boundedness of smooth operator-valued functions*, Integral Equations Operator Theory 63 (2009), 373–402; arXiv:0804.3313.
- [2] J. Lindenstrauss and L. Tzafriri, *Classical Banach Spaces II: Function Spaces*, Springer, 1979.